Akka vs. n8n

A comparison for teams building agentic AI

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n8n automates your workflows; the Akka Agentic AI Platform runs your production agentic systems — and guarantees resilience and scalability. n8n is excellent visual workflow automation with AI Agent nodes — the fast path for integrations and internal automations. It is not built for production agentic scale or durable execution, and its AI guardrails and approvals cover workflows and agents running in n8n: the heavy-load, reliability, and compliance burden stays with you.

99.5%	99.9999%	4ms	5B
N8N ENTERPRISE SLA	AKKA SLA	AKKA STATE READS	TOKENS / SEC BENCHMARK

DIMENSION	N8N	AKKA
What it is	A visual workflow-automation platform with AI Agent nodes	The Akka Agentic AI Platform — guarantees resilience and scalability
Scope	Workflow automation and integrations, plus n8n Agents (Preview); agentic scale and durable execution are the customer's burden	Orchestration, agents, memory, streaming, APIs, observability, and governance on one runtime
Availability SLA	99.5% uptime (Enterprise Cloud), excluding planned maintenance	99.9999% — entire platform, backed by indemnities
HA / DR	Queue mode + multi-main, Redis required; no published RTO/RPO or active-active multi-region guarantee	Sub-1-minute RTO; zero-byte RPO; active-active across regions
Durable execution	Execution state persisted to PostgreSQL; docs describe a 30-second graceful worker shutdown and no replay of interrupted runs	Event-sourced; the runtime handles recovery and replay
AI agents	AI Agent node built on the LangChain JS framework; n8n Agents (Preview), not yet in queue mode	Native agents with tools, handoffs, guardrails, interaction logging
Memory	n8n Agents: built-in session memory; AI Agent node: sub-nodes, external store in queue mode	Durable in-memory, 4ms reads / sub-10ms writes
Governance / EU AI Act	Guardrails node, human tool approval, node and credential policies; audit logs (Enterprise); no named AI regulation	Aspect-woven runtime enforcement + full pre-production governance
Cost model	Per-execution metering (€20–€667/mo billed annually; Enterprise custom) + infra you operate when self-hosted	Shared compute; up to 90% lower infrastructure for the same workload
Certifications	SOC 2 (annual audit; report to enterprise customers), public SOC 3, GDPR	19 standards (SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF)
License	Sustainable Use License (fair-code, source-available, not open source); internal-business-purpose only; reselling n8n as a service prohibited; n8n Embed needs a commercial license	BSL; deploy anywhere including sovereign cloud; portable specs

n8n Is an Automation Tool; Akka Is an Agentic Platform

n8n is a visual workflow-automation platform that connects 2,000+ apps and APIs, with AI Agent nodes and n8n Agents (Preview) added on top. It is excellent for integrations and internal automations, and self-hosting gives teams full control of their data. A production agentic system needs more than node-based automation — durable execution, horizontal agentic scale, streaming, and governance across every AI system — and n8n leaves those to you.

Capability	n8n	Akka
Visual workflow automation + integrations	Yes (2,000+ integrations)	Programmatic; integrations via code
Native AI agents (tools, handoffs, streaming)	AI Agent node on LangChain JS; n8n Agents (Preview)	Built in
Durable execution / replay	PostgreSQL-persisted; no documented replay of interrupted runs	Event-sourced, runtime-managed recovery
Durable memory	Session memory in n8n Agents (Preview); sub-nodes in workflows	Built in, 4ms / sub-10ms
Real-time streaming	None native	Built in, backpressured, petabyte-scale
Inline governance / policy enforcement	Guardrails node; human tool approval; node and credential policies	Inline, runtime-embedded
Pre-production governance	Workflow reviews; policies that refuse a publish	Classification, sign-offs, sealed posture

n8n's AI Agent node is a wrapper over the LangChain JavaScript framework. It is ideal for triggering an LLM step inside an automation; it is not an agent runtime engineered for unattended, process-attached execution at scale. n8n Agents (Preview, August 2026) add standalone agents with session memory, tools, and Slack, Telegram, and Linear channels; they do not yet run in queue mode.

Everything an agentic system needs runs in one Akka runtime, so a system that compiles is ready to run in production. The Akka SDK models the entire agentic system in one place — agents, tools, durable memory, streaming, APIs, and governance — and the platform generates it, tests it, and runs it as one certified service.

Reliability, Scale, and Durable Execution

n8n Enterprise Cloud publishes a 99.5% uptime commitment (excluding planned maintenance during 10am–5pm German time) — roughly 1.8 days of allowed downtime per year. The Akka contractual SLA is 99.9999% — about 31 seconds per year — across the entire platform, backed by indemnities.

Metric	n8n	Akka
Availability SLA	99.5% (Enterprise Cloud)	99.9999%
Allowed downtime / year	~1.8 days	~31 seconds
RTO	Not published	Sub-1 minute
RPO	Not published	Zero byte
HA model	Queue mode + multi-main; Redis + PostgreSQL required	Active-active across regions
Durable execution	PostgreSQL-persisted; no documented replay of interrupted runs	Event-sourced, replayable from the journal

To scale horizontally, n8n runs in **queue mode**: a Redis message broker (Bull queue) distributes executions to worker processes, PostgreSQL is mandatory, and multiple "main" instances provide availability. This is capable for automation throughput. On shutdown a worker gets 30 seconds by default to finish its jobs, and the queue-mode documentation describes no replay of a run interrupted mid-step. The Akka actor-based runtime runs 4KB actors at 200M actors/core across clusters up to 1M nodes; state is durable in-memory and replayable from the event journal; the runtime owns recovery, so application code does not have to.

Continuous AI Intelligence

Akka Optimize keeps improving the AI after it ships. It runs an evaluation loop on your own production traffic, grading live interactions and training smaller specialized models on your proprietary data, so the system keeps getting better as it runs. The loop delivers three results that compound over time:

- **Model leverage** — route each request to the best model from any vendor under your policies, and reserve frontier models for the requests that need them.
- **Data sovereignty** — the specialized models are trained on your proprietary data, owned by you, and run inside your own environment.
- **Cost governance** — full visibility and control of AI spend, with savings that compound as the loop keeps running.

The evaluations run continuously inside Akka and grade traffic from agents wherever they run, in Akka or in third-party harnesses.

Up to 90% Cheaper to Operate

AI systems built with Akka are up to **90% cheaper to operate** than Python-based systems — a function of the infrastructure required for the same agentic transaction volume, not list price. The drivers are actor concurrency (~10T tokens/core/year vs ~2T; ~80% less compute than Python frameworks; Manulife: up to 300% more concurrency, 30–50% faster processing after porting from Python), shared compute, and micro-checkpointing.

n8n bills **per execution** (€20/mo Starter / 2,500 executions, up to €667/mo Business / 40,000 executions, billed annually; Enterprise custom), and each agent turn counts as one execution. Streaming and the database sit outside the meter — provisioned and operated by you when self-hosted. Akka runs all of it on one shared-compute runtime for a fixed annual fee finance can forecast.

Governance and the EU AI Act

The n8n Guardrails node blocks or redacts text that fails a jailbreak, PII, secret-key, topic, or custom check. An AI Agent can pause a tool call until a person approves it, and instance and project policies refuse a save, publish, or run that uses a blocked node or credential type. The controls cover workflows and agents running in n8n. Enterprise adds audit logs and log streaming. n8n maps its controls to no AI regulation such as the EU AI Act.

Violation	Maximum Fine
Prohibited AI practices (Art. 5)	EUR 35M or 7% global turnover
High-risk obligations (Art. 9-15)	EUR 15M or 3% global turnover

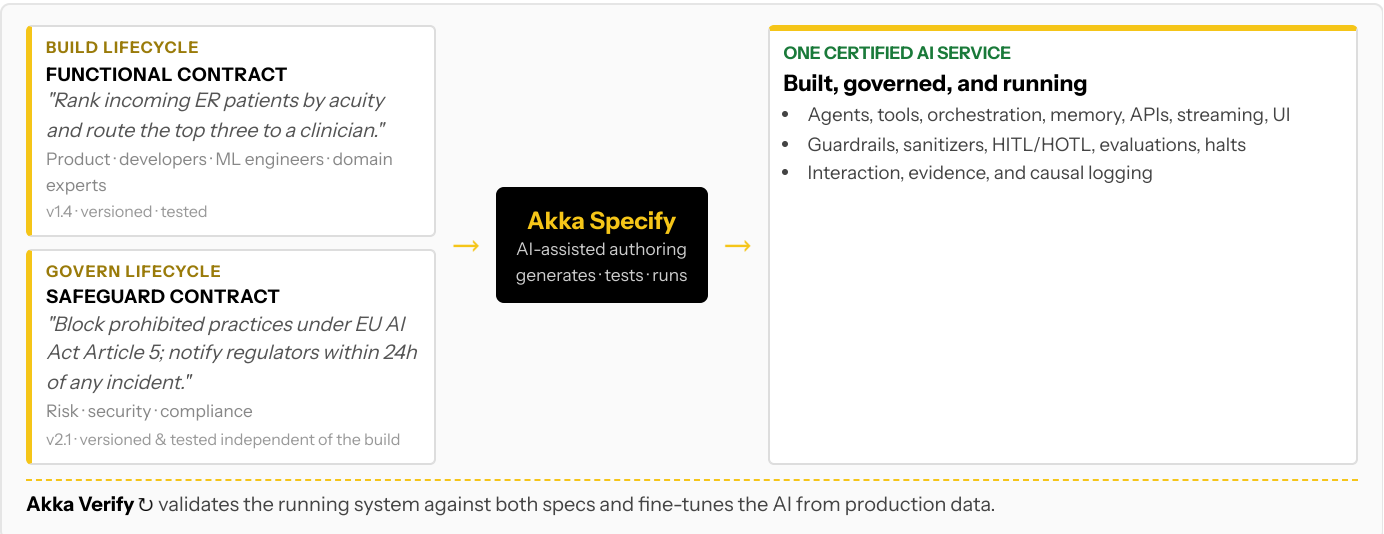
High-risk AI carries a 10-year logging-retention obligation (Art. 72).

How Akka Governs Inline

The EU AI Act is enforceable now (prohibited practices since Feb 2025, high-risk since Aug 2025). Akka governs at the runtime: inline guardrails, policies, LLMs-as-a-judge, and sanitizers; hash-chained immutable evidence; HITL/HOTL control; pre-deployment classification against **241 regulations and frameworks and 1,718 controls** (699 of which carry a financial penalty); multi-persona sign-offs; a sealed Governance Posture Package; and Akka Verify proving conformance from the running system. Controls n8n applies inside its own workflows, Akka Verify applies to any AI system.

Two Lifecycles, One Certified System

Building on n8n means a workflow author wiring nodes on a canvas, with admin policies on node and credential types and review requests before changes publish; there is no co-equal governance lifecycle that risk and compliance author and version independently. Akka runs two independent lifecycles on one platform via **Akka Specify**:



The build lifecycle and the governance lifecycle are versioned and tested independently, by different audiences — an audience and a workflow n8n has no equivalent for.

Real-Time Streaming at Petabyte Scale

n8n has no native streaming engine; real-time pipelines and agent feedback loops are provisioned separately. Akka streaming is built into the runtime — continuous, backpressured, **petabyte-scale, in-memory**, with no external broker — powering both agent feedback loops and high-throughput data processing (the engine behind Tubi's real-time hyper-personalization at 5 billion tokens per second).

For the Buyer: Risk, Compliance, Accountability, and Licensing

Buyer concern	n8n	Akka
Certifications & audits	SOC 2 (annual independent audit; report for enterprise customers), public SOC 3, GDPR (n8n.io/legal/security)	19 standards — SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF — plus annual pen tests, SBOMs, 40+ policies (trust.akka.io)
Scope of accountability	The automation layer; you operate scale, durability, and governance beyond n8n workflows	One platform, one SLA, 24/7 SRE — Akka owns the running system
Risk transfer	Standard cloud terms	Availability and data-integrity guarantees backed by contractual indemnities
Track record & funding model	Venture-funded: $180M Series C at a $2.5B valuation (Oct 2025); SAP strategic investment at a $5.2B valuation (May 2026); founded 2019; scaling on investor capital	Profitable and self-funding; since 2009 and 100,000+ deployments (52 banks); Dell Technologies Capital is largest shareholder, a customer, and an AI partner
Budget predictability	Per-execution metering that scales with load	Fixed annual fee finance can forecast
License	Sustainable Use License — fair-code, source-available, not open source; internal-business-purpose only; reselling/hosting n8n as a service prohibited; embedding requires the n8n Embed commercial license	BSL; deploy on Akka cloud, hyperscaler VPC, your Kubernetes, on-prem, or sovereign cloud; portable specs

n8n is well-funded and growing fast — the decision is scope and accountability. n8n gives you a best-in-class automation canvas to integrate apps and trigger AI steps; Akka gives you the agentic platform, and owns the SLA, the durability, and the governance. The Sustainable Use License is also a procurement gate worth reading early: source-available but commercially restricted, with no right to offer n8n as a hosted service and a separate commercial license required to embed it.

Customers Running Agentic and Real-Time Systems on Akka

Manulife
2,000
developers across 100 projects on one governed platform

Tubi
5B tok/s
real-time hyper-personalization engine

Swiggy
71ms
order-assignment AI, ~50% latency reduction

John Deere
1,000+
tractor sensors turned into real-time insight

Verizon
750%
order-processing capacity gain; 6s → 2.4s response

Common Questions

We already run n8n for our automations. Why add Akka?

Keep n8n for what it is excellent at — integrations and internal automations. When you move into production agentic AI you also need durable execution, horizontal agentic scale, streaming, and governance across every AI system, which n8n does not provide. Akka runs the agentic systems on one platform while n8n keeps handling your automations.

n8n has AI agents and SAP invested at a $5.2B valuation. Doesn't that close the gap?

n8n's AI Agent node is a LangChain-based step inside a workflow, and n8n Agents (Preview) do not yet run in queue mode. Strong funding reflects real demand for automation. n8n guardrails and approvals cover workflows and agents running in n8n, and Enterprise Cloud

Can we add governance on top of n8n?

n8n already provides the Guardrails node, human approval of tool calls, admin policies on node and credential types, and audit logs with log streaming on Enterprise. Those controls cover workflows and agents running in n8n and map to no named regulation. The EU AI Act expects immutable records witnessed as they happen, classification before a system ships, and a sealed audit artifact. Akka Verify applies any control to any AI system, guarantees its enforcement, and covers pre-deployment governance.

Isn't n8n cheaper because it is free to self-host?

The n8n engine is source-available under the Sustainable Use License (fair-code, not open source), and self-hosting means you build and operate scale, HA/DR, durability, and everything around it. Cloud bills per execution. The Akka shared-compute model is up to 90% cheaper

commits to 99.5% uptime. Capitalization does not add a platform-wide six-nines SLA or governance across every AI system.

to operate than the equivalent assembled stack, on a fixed annual fee — and the license is a procurement gate: reselling n8n as a service is prohibited and embedding requires a commercial license.

Sources

n8n Enterprise Cloud SLA: n8n.io/legal/cloud-enterprise-terms — 99.5% uptime, excluding planned maintenance (10am–5pm German time)

n8n scaling / queue mode: docs.n8n.io/hosting/scaling/queue-mode — Redis (Bull queue) + worker processes, PostgreSQL required, multi-main for availability

n8n durable execution: docs.n8n.io/hosting/scaling/queue-mode — 30-second graceful worker shutdown; no documented replay of interrupted runs; community.n8n.io & github.com/n8n-io/n8n #22541 report runs marked "crashed"

n8n AI controls & Agents: n8n.io/integrations, docs.n8n.io guardrails node, human-in-the-loop tools, build-and-manage-agents, release notes 2.42–2.44 — 2,000+ integrations; inline guardrails, tool approval, node and credential policies, agent session memory

n8n AI Agent node: docs.n8n.io/integrations/builtin/cluster-nodes/root-nodes/n8n-nodes-langchain.agent — built on the LangChain JS framework

n8n pricing: n8n.io/pricing — per-execution metering, €20/mo Starter (2,500 exec) to €667/mo Business (40,000 exec), billed annually; Enterprise custom

n8n license: docs.n8n.io/sustainable-use-license & blog.n8n.io/announcing-new-sustainable-use-license — fair-code, source-available, not open source; internal use only; no reselling as a service; n8n Embed commercial license

n8n security & certifications: n8n.io/legal/security & trust.n8n.io — SOC 2 (annual independent audit; report for enterprise customers), public SOC 3, GDPR

n8n funding: blog.n8n.io/series-c — $180M Series C at $2.5B valuation (Oct 2025); SAP strategic investment at $5.2B valuation (May 12, 2026); founded 2019

Akka performance: akka.io/blog/go-slow-to-go-fast — Manulife up to 300% more concurrency, 30–50% faster; ~10T vs ~2T tokens/core; ~80% less compute than Python

Akka trust center: trust.akka.io — 19 compliance standards; SOC 2 II + public SOC 3; annual pen tests, SBOMs, 40+ policies

Akka platform: 99.9999% availability, active-active HA/DR, sub-1 min RTO, zero byte RPO (contractual indemnities); 241 regulations and frameworks / 1,718 controls / 699 with a financial penalty; 100,000+ deployments / since 2009; profitable; Dell Technologies Capital